

OpenMind(Steam): Query-Time Diversification via Logical Rules

This survey only takes you less than 15 minutes to complete.

IMPORTANT: Please read the instruction carefully and select your answers. We will check the consistency of your answers to make sure that you do not answer the questions by random selection.

CAUTION: If you provide inconsistent answers, your result will get **REJECTED!**

We would like to do the pre-survey demographics questions before formal questions.

* Indicates required question

1. **1 . What is your age group? ***

Check all that apply.

- ☐ Under 18
- ☐ 18-35
- ☐ 35-44
- ☐ 45-54
- ☐ 55 or above

2. **2. What is your highest level of education? ***

Check all that apply.

- ☐ High school or below
- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ Doctoral degree or above

3. **3. Do you have any prior experience with personalized recommendation systems (e.g., product, video, or music recommendations)? ***

Check all that apply.

- ☐ Yes
- ☐ No
- ☐ Not sure

Preliminary (Game recommendation)

We will consider a **game recommendation** scenario, which **recommends some games to users**.

We aim to provide **diversified top-5 recommendations for user to break information cocoon** caused by the traditional recommender system.

We will give the **historical interactions of a user u_0** .

Based on such information, we will show you **FOUR top-5 recommended lists to user u_0** such that user u_0 can be exposed to the game with different viewpoints/perspectives which can be easily accepted by user u_0 , yielding to **break information cocoon** caused by the traditional recommender system.

4. His Interaction History

We provide his interaction history with games in below.

Each game is provided some information, including the description and the keywords (topics).

Interaction History	Description	Keywords (Topics)
Cold Fear	During a severe Arctic storm, a coast-guard veteran boards a mysterious Russian whaler and discovers horrors inside.	Horror; Investigation; Survival
Crysis 2	Aliens invade New York, and the player uses advanced equipment to fight through the city and survive.	Science fiction; Shooting; Action
Deadlight	After a catastrophe, a city is occupied by dangerous infected people; the player runs, jumps, and searches for safe routes.	Zombies; Platforming; Survival
ABZU	The player dives into a colorful underwater world, exploring the ocean and discovering its living creatures.	Ocean; Relaxing; Exploration
Layers of Fear	The player is a painter exploring a changing Victorian mansion while facing disturbing psychological visions.	Psychological horror; Exploration; Story
Saints Row IV	The player leads a group against an alien invasion, using unusual abilities and weapons across a city.	Action; Science; City exploration;
Vanquish	This science-fiction action game has the player fight future-tech enemies with a wide range of weapons.	Science fiction; Shooting; Fast
Agents of Mayhem	The player commands a special-operations team in Seoul against a criminal organization trying to control the world.	Action; City; Team-based play

Short User Preference

Overall, this player mainly enjoys **science-fiction action and shooter games**, with a strong interest in **horror, survival, and exploration**, and occasional openness to **relaxing, story-driven experiences**.

5. Traditional Top-5 Recommendation list

It contains five games, namely, **IT Simulator / Penumbra: Black Plague Gold Edition / Montague's Mount / Conarium / Decay: The Mare**.

Short description:

These games are largely similar in tone and theme.

All center on **horror and psychological tension**, sharing a dark, unsettling atmosphere built around exploration and puzzle-solving.

Game	Description	Keywords (Topics)
IT Simulator	Two players cooperate to restore power to an office building, taking the roles of an all-seeing control-room operator and a stealthy repair technician trying to escape.	Cooperation; Horror; Stealth
Penumbra: Black Plague Gold Edition	The player gathers clues and solves puzzles while surrounded by enemies and psychological terror that threatens the protagonist's sanity.	Psychological horror; Puzzles; Survival
Montague's Mount	The player explores a bleak, abandoned Irish island, searching the environment and solving challenging puzzles in a psychological thriller.	Psychological thriller; Island exploration; Puzzles
Conarium	The player follows a story about four scientists and explores the strange aftermath of their attempt to challenge the accepted limits of nature.	Horror; Scientific mystery; Exploration
Decay: The Mare	A man entering a treatment institution becomes trapped in an endless nightmare on his first night, and the player explores the setting to find a way out.	Psychological horror; Nightmare; Escape

Objective

We would like to provide diversified recommendations for this user to break this information cocoon.

The diversified recommended item should be easily accepted by user and at the same time, provide alternative viewpoints/perspectives with respect to the traditionally recommended games.

Questions

Consider the following **four top-5 recommendation lists** for making the user u_0 exposed to diverse items to broaden his views or perspectives.

You will be asked **eight questions** based on the four top-5 lists.

Recommendation 1

It contains five games, namely, **Blair Witch / F.E.A.R. / ZOMBI / DARQ / State of Decay**.

Short description:

These games are **largely similar** in tone and theme.

All center on **horror and survival**, sharing a dark, tense atmosphere.

Within that focus there is some mild complementarity:

(a) **F.E.A.R.** leans toward shooting, (b) **DARQ** toward puzzle-solving, and (c) **State of Decay** toward base-building and resource management.

But the overall experience is unified around **fear and survival** rather than offering distinct directions.

Game	Description	Keywords (Topics)
Blair Witch	The player investigates a disappearance in a forest while facing strange events that affect perception and fear.	Forest; Investigation; Psychological horror;
F.E.A.R.	The player investigates supernatural events while fighting with guns in dark and frightening environments.	Horror; Shooting; Supernatural
ZOMBI	A plague has overwhelmed London; the player searches for scarce supplies and avoids infected people.	Zombies; Survival; Resources
DARQ: Complete Edition	A boy is trapped in a lucid nightmare and must observe the environment, solve puzzles, and find a way out.	Dreams; Puzzles; Horror
State of Decay	After a zombie disaster, the player gathers supplies, helps survivors, and builds a safer base.	Zombies; Survivors; Base building

Recommendation 2:

It contains five games, namely, **Half-Life: Blue Shift / Half-Life: Caged / Half-Life 2: Episode Two / Tomb Raider / Redneck Rampage**.

Short description:

These games are **similar**, built around **shooting and sci-fi action**.

Most share the same core loop of **combat, escape, and exploration**, and three of them come from the same Half-Life universe.

The list offers little variety, the titles reinforce one another rather than covering different play styles.

Game	Description	Keywords (Topics)
Half-Life: Blue Shift	The player is a security guard in a research facility who seeks a way out after a disaster and fights enemies.	Science fiction; Shooting; Escape
Half-Life: Caged	The player is trapped in a prison-like setting and must fight, explore, and find a way to escape.	Shooting; Escape; Exploration
Half-Life 2: Episode Two	The player continues through an occupied city and wilderness, working with companions against hostile forces.	Science fiction; Adventure; Combat
Tomb Raider	The player follows Lara as she explores ruins, climbs through environments, solves puzzles, and fights enemies.	Exploration; Puzzles; Action
Redneck Rampage	The player uses weapons against aliens and other enemies in rural and wilderness settings.	Action; Shooting; Wilderness

Recommendation 3:

It contains five games, namely, **Sisters Royale / Monster Boy and the Cursed Kingdom / Golf Gang / Teeworlds / CHRONO TRIGGER.**

Short description:

These games are **complementary**.

Each fills a different niche: **arcade shooting, fantasy platforming, casual sports, multiplayer battles, and story-rich role-playing.**

Rather than repeating one theme, they cover a wide spread of genres and moods, giving the player varied and mutually distinct options.

Game	Description	Keywords (Topics)
Sisters Royale: Five Sisters Under Fire	Five sisters compete over the same goal in a fast vertical shooting game that emphasizes aiming and avoiding attacks.	Shooting; Arcade; Sisters
Monster Boy and the Cursed Kingdom	The player adventures through a colorful fantasy world, fights enemies, and uses different forms to solve environmental problems.	Fantasy; Adventure; Platforming
Golf Gang	Players compete in fast mini-golf rounds, alone or online, trying to complete courses quickly.	Sports; Racing; Multiplayer
Teeworlds	Players use weapons in simple multiplayer battles, including team matches, and can also design their own maps.	Multiplayer; Battles; Map design
CHRONO TRIGGER	The player travels across different historical eras, adventures with companions, and tries to save the world.	Role-playing; Time travel; Adventure

Recommendation 4

It contains five games, namely, **Tomb Raider / Life is Strange – Episode 1 / Batman: Arkham City / Rise of the Tomb Raider / Shadow of the Tomb Raider**.

Short description:

These games are **mostly similar**, centered on **exploration, action, and puzzle-solving**.

Three Tomb Raider titles share nearly identical gameplay, and Batman adds a comparable action-investigation style.

Life is Strange is the one **complementary** outlier, offering a slower, narrative- and choice-driven experience that contrasts with the action focus of the rest.

Game	Description	Keywords (Topics)
Tomb Raider	The player follows Lara as she explores ruins, climbs through environments, solves puzzles, and fights enemies.	Exploration; Puzzles; Action
Life is Strange - Episode 1	The player explores a school and a small town, using the ability to change the past to affect the story.	Story; Choices; Time
Batman: Arkham City - Game of the Year Edition	The player controls Batman, investigating events, moving quietly, and fighting enemies across a city.	Hero; Action; Investigation
Rise of the Tomb Raider	The player follows Lara through wilderness and ancient ruins, solving puzzles and dealing with danger.	Exploration; Ruins; Survival
Shadow of the Tomb Raider: Definitive Edition	Lara tries to stop a Maya apocalypse while exploring ruins and varied environments during her journey.	Exploration; Action; Ancient ruins

Evaluation

Consider the quality of the above four recommendations for breaking information cocoon. In particular, focus on the following four aspects of each recommendation with respect to user history:

1. Acceptability
2. Topic Consistency
3. View Divergence
4. Overall Performance

Acceptability

"Acceptability" is an aspect to be evaluated in this survey.

A given top-5 recommendation list is considered as good in terms of acceptability **if it can be easily accepted by the user u_0** .

Thus, a recommendation list with higher acceptability is better.

6. Q1. Consider only "**Acceptability**" of each top-5 recommendation list. Please indicate one recommendation that you think with the best "Acceptability" in all four diversified top-5 recommendation lists. *

Mark only one oval.

- ☐ Recommendation 1
- ☐ Recommendation 2
- ☐ Recommendation 3
- ☐ Recommendation 4

7. Q2. Consider only "Acceptability" of each top-5 recommendation list. Please indicate **exactly 2 recommendations** that you think with better "Acceptability" in all four top-5 recommendation lists. Note: One of them **must** be the top-5 recommendation list you indicated in Q1. *

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

Topic Consistency

"Topic Consistency" is an aspect to be evaluated in this survey.

A given top-5 recommendation list is considered as good in terms of topic consistency **if it maintains consistent topic with the traditional recommendation.**

Thus, a top-5 recommendation list with higher consistency is better.

8. Q3. Consider only "**Topic Consistency**" of each top-5 recommendation list. *
Please indicate one top-5 recommendation list that you think with the best "Topic Consistency".

Mark only one oval.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

9. Q4. Consider only "**Topic Consistency**" of each top-5 recommendation list. Please indicate **exactly 2 top-5 recommendation lists** that you think with better "Topic Consistency" in all four top-5 recommendation lists. Note: One of them **must** be the top-5 recommendation list you indicated in Q3. *

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

View Divergence

"View Divergence" is an aspect to be evaluated in this survey.

View Divergence measures whether the top-5 recommendation list provides a **different perspective or worldview** from the reference work, rather than simply a different genre or topic.

A high-scoring top-5 recommendation list should differ in playing settings/details, while remaining comparable at the content level.

Therefore, **genre/theme difference alone should not be considered sufficient** for view divergence.

Thus, a top-5 recommendation list that provides different viewpoint/perspectives is better than one that falls the same flow of playing settings.

10. Q5. Consider only "**View Divergence**" of each top-5 recommendation list. Please indicate one top-5 recommendation list that you think with the best "View Divergence". *

Mark only one oval.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

11. Q6. Consider only "**View Divergence**" of each top-5 recommendation list. Please indicate *exactly 2 top-5 recommendation lists* that you think with better "View Divergence" in all four top-5 recommendation lists. Note: One of them *must* be the top-5 recommendation list you indicated in Q5. *

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

Overall Performance

"Overall Performance" is the last aspect to be evaluated in this survey.

It is an aspect of the overall performance of each top-5 recommendation list.

Under the aspect of the overall performance,

we would like to have **a top-5 recommendation list such that it is good in terms of the acceptability, topic consistency and view divergency simultaneously.**

In this case, the user u_0 can accept this recommendation with high probability, broadening his horizons,

and then, breaking the information cocoon caused by the traditional recommender system.

Thus, if the performance of a top-5 recommendation list is better than another one in terms of the acceptability, topic consistency and view divergency, the former recommendation is better.

12. Q7. Consider only "Overall Performance" of each top-5 recommendation list. Please indicate one top-5 recommendation list that you think with the best "Overall Performance". *

Mark only one oval.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

13. Q8. Consider only "Overall Performance" of each top-5 recommendation list. Please indicate *exactly 2 top-5 recommendation lists* that you think with better "Overall Performance" in all four top-5 recommendation lists. Note: One of them *must* be the top-5 recommendation list you indicated in Q7. *

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

Thank you so much!

Thank you so much for your help!

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